Topics SWA midterm exam

Lesson 1: SWA overview

- What is software architecture
- Difference enterprise architecture and software architecture
- Difference architecture in a waterfall and agile project
- Characteristics of software qualities
- Architecture principles

Lesson 2: Domain Driven Design

- Principles of DDD
- Anemic and rich domain model
- Orchestration and choreography
- DDD patterns: entity, value object (and its characteristics), domain service and domain event

You need to be able to design a rich domain model in UML where you specify for each domain class what DDD type it is

Lesson 3: Databases

- Scaling databases
    - Horizontal, vertical scaling
    - Sharding/partitioning
    - replication
    - Brewers CAP theorem
- Relational databases
    - Characteristics, Advantages and disadvantages
    - Problems of relational databases
- Key-value store(Redis)
    - Characteristics, Advantages and disadvantages
- Document store (mongodb)
    - Characteristics, Advantages and disadvantages
- Column family store (cassandra)
    - Characteristics, Advantages and disadvantages
- Graph database (neo4j)
    - Characteristics, Advantages and disadvantages
- When do you choose which database
- How is the data structured in each of these databases

You do not need to write spring boot code for database access, but you do need to understand given spring boot code for database access.

Lesson 4: Component based design

- What are components
- Why do we need components
- API design best practices
- Component design
- DTO's

You need to be able to design one or more components in UML
You need to be able to write a good interface
You need to understand how to implement a component based monolith.

Lesson 5: SOA and spring integration

- Hub and spoke
  - Characteristics, advantages, disadvantages
- SOA, ESB
  - Characteristics, advantages, disadvantages
- Integration patterns

You need to understand all given integration patterns. You do not need to draw the given pictures, but you need to understand them.

- Spring integration

You need to understand the concepts of how spring integration works.
You do not need to write XML configuration, but you need to understand a given XML file.
You do need to know how to write Java implementation of the following patterns:

- Custom router
- Filter

<u>Lesson 6</u>

Monolith: advantages and disadvantages
Characteristics of a microservice
Advantages and disadvantages of a microservice
Microservice and database
Microservice and UI, micro-front-ends
Microservice boundaries
Domains: core, supporting and generic
Microservice in the organization:
devops team
Conways law

Feign
Registry
You do not need to write feign or registry code or configuration. You do need to understand the core concepts and principles of these techniques.

Lesson 7

Registry failover (replicas)
Load balancing :client side, server side
API gateway
You do not need to write Java code or configuration code yourself.
You do need to understand the core principles and concepts of these techniques.